Improving variable selection properties with data integration and transfer learning

Abstract

We study variable selection (also called support recovery) in high-dimensional sparse linear regression when one has external information on which variables are likely to be associated with the response. Consistent recovery is only possible under somewhat restrictive conditions on sample size, dimension, signal strength, and sparsity. We investigate how these conditions can be relaxed by incorporating said external information. A key application that we consider is structural transfer learning, where variables selected in one or more source datasets are used to guide variable selection in a target dataset. We introduce a family of likelihood penalties that depend on the external information, motivated by connections to Bayesian variable selection. We show that these methods achieve variable selection consistency in regimes where any method ignoring external information fails, and that they achieve consistency at faster rates. We first quantify the potential gains under ideal, oracle-chosen, penalties. We then propose computationally efficient empirical Bayes procedures that learn suitable penalties from the data. We prove that these procedures have improved variable selection properties compared to methods that do not use external information. We illustrate our approach using simulations and a genomics application, where results from mouse experiments are used to inform variable selection for gene expression data in humans.

Keywords: high-dimensional statistics, Bayesian variable selection, ℓ_0 penalty, empirical Bayes, transfer learning.

1 Introduction

Many contemporary statistical problems involve high-dimensional models, where the goal is to identify a small subset of variables associated with a response. In such settings, inference is often severely limited by the dataset sample size. At the same time, basing inference on a single dataset increasingly feels like a wasted opportunity: it is common to have additional external information—such as prior studies, auxiliary datasets, or variable-specific annotations—that can suggest differential relevance across variables. This situation arises, for example, in multimodal data (Liang et al. 2023) where different groups of variables may exhibit different levels of sparsity, such as clinical versus genomic markers in biomedical applications. It also arises when metadata are available on the variables themselves, for instance through functional annotations of genes.

A setting of particular interest is transfer learning, which broadly refers to improving performance on a target dataset by leveraging information from related source datasets. In the context of variable selection, transfer learning uses information about variables identified in source datasets as external input to guide variable selection in a target regression problem. We refer to this problem as *structural transfer learning*, to distinguish it from approaches that focus on transferring parameter estimates or predictions.

In high-dimensional settings, leveraging external information is particularly attractive. From a theoretical perspective, the limited information provided by a single dataset implies that strong assumptions are required for accurate inference. In structural learning problems such as variable selection, those assumptions regard sparsity and signal strength and, although mathematically necessary, they can be too strong in practice (see Giannone et al. (2021) for a discussion in the context of social sciences). From a practical perspective, numerous works observed improved inference when employing external information to guide structural learning (Stingo et al. 2011, Boulesteix et al. 2017, Chen et al. 2021).

Despite this empirical success, a theoretical framework explaining precisely why and how leveraging external information may deliver improved structural learning is not currently available. Our main contribution is investigating how, in linear regression, integrating external information (and structural transfer learning as a particular case) would allow an oracle to push the mathematical conditions under which consistent variable selection is possible, and improving the corresponding rates. Consistency is understood as recovering the truly active variables with probability going to 1 as the sample size n, and possibly the dimension p, go to infinity. Our other main contribution is proposing an empirical Bayes data-based procedure that doesn't require oracle information, and showing that it also enjoys improved properties.

To make ideas concrete, consider the Gaussian linear regression

$$\mathbf{y} \;=\; \mathbf{X}\boldsymbol{\beta}^* + \epsilon, \qquad \epsilon \sim N(0, \sigma^2 I_n), \tag{1}$$

where $\mathbf{X} \in \mathbb{R}^{n\times p}$, $\mathbf{y} \in \mathbb{R}^n$, $\sigma > 0$, and $\boldsymbol{\beta}^* \in \mathbb{R}^p$ are the data-generating parameters. Our results can be easily extended to non-linear regression, where $\mathbf{X}$ is given by a suitable basis such as splines. Penalized likelihood methods for variable selection are often based on optimizing the log-likelihood plus a penalty term driven by the ℓ_q "norm" of an estimated $\hat{\boldsymbol{\beta}}$ for $q \in [0, 1]$. Here we focus on ℓ_0 penalties because they possess theoretically superior properties for support recovery compared to other penalties (Wainwright 2010, Gao & Aragam 2025). That is, our goal is to study how external information can improve upon these best known properties.

Besides, computational advances have made ℓ_0 problems much more tractable: optimization methods can solve them exactly for p in the thousands (Bertsimas & Van Parys 2020) and Markov Chain Monte Carlo (MCMC) methods have linear cost in p under sparsity conditions (Yang et al. 2016, Zhou et al. 2022). There is also a direct connection between ℓ_0 penalties and Bayesian variable selection (Schwarz 1978, Chen & Chen 2008, Rossell

2022). Under mild regularity conditions, a choice of ℓ_0 penalty corresponds to a choice of prior probability for variable inclusion. This connection converts ℓ_0 penalization into a natural and intuitive way to encode external information on the relevance of variables for association with the outcome.

We consider a variable selection framework where ℓ_0 penalties may depend on external information $\boldsymbol{z} = (z_1, \ldots, z_p) \in \mathbb{R}^p$. In transfer learning applications, each z_j may be a q-dimensional vector measuring the importance of variable j in q source datasets. For theoretical tractability and to be able to provide sharp conditions and rates, here we focus on the case where $\boldsymbol{z}$ partitions the p variables into b blocks, that is $z_j \in \{1, \ldots, b\}$. We denote by $B_j \subset \{1, \ldots, p\}$ the set of variables in block$j = 1, \ldots, b$. When $\boldsymbol{z}$ is informative for variable selection, key characteristics such as the proportion of truly active variables or the signal strength may vary across blocks, hence one allows the ℓ_0 penalty to differ across blocks. We derive the variable selection properties of such informed ℓ_0 penalties, first in general and then for structural transfer learning, and compare them to standard ℓ_0 penalties.

Before proceeding, we review selected literature. External information has been used to guide variable selection in many applications. For instance, Stingo et al. (2011) and Cassese et al. (2014) proposed Bayesian variable selection for gene expression where prior inclusion probabilities depend on biological knowledge and meta-variables. Chen et al. (2021) predicted disease outcomes with ℓ_1 penalties that depend on gene functional annotations. Several works discussed methodology, but not theory, for external-information dependent penalization or prior inclusion probabilities. In the Bayesian literature, van de Wiel et al. (2019) suggested an empirical Bayes approach to regress prior inclusion probabilities on external data. Velten & Huber (2021) proposed a variational Bayes approximation for regression with external information-dependent prior inclusion probabilities. Recently, Busatto & van de Wiel (2023) considered a horseshoe prior that depends on external information. In the frequentist literature, in the context of multi-omics data, Boulesteix et al. (2017) let the ℓ_1 penalty depend on the modality of each variable. Zeng et al. (2020) and van Nee et al. (2023) proposed empirical Bayes to set externally-informed ℓ_1 and elastic-net penalties respectively. See van de Wiel & van Wieringen (2024) for a review.

Previous theory on transfer learning in regression has focused on parameter estimation, but provided no results on identifying the set of active variables as we do here. Jiang et al. (2016) and Li et al. (2021) showed improvements in convergence rates in prediction and estimation with ℓ_1 penalties that depend on the discrepancy with source estimates. In a Bayesian setting, Lai et al. (2024) showed improvements in posterior contraction rates of parameters with horseshoe priors centered at a weighted average of source-specific estimates.

The paper is organized as follows. In Section 2, we define externally-informed ℓ_0 penalties and discuss their connection to Bayesian variable selection. Section 3 studies their theoretical properties, including oracle results characterizing the maximum gains achievable by informed ℓ_0 penalties. In Section 4, we propose a data-driven empirical Bayes procedure for setting informed ℓ_0 penalties. Section 5 applies this procedure to transfer learning and establishes improved selection properties, including robustness to negative transfer in settings where the external data is uninformative regarding what variables are truly active, and to covariate shifts. Section 6 shows simulations and a gene expression study transferring findings from mice to humans.

Notation: The parameters of interest in the target regression are denoted by $\boldsymbol{\beta} \in \mathbb{R}^p$ and $\boldsymbol{\beta}^*$ denotes their true values. The set of variable indices is $V = \{1, \ldots, p\}$. For any $A \subseteq V$ and any vector $\mathbf{x} \in \mathbb{R}^p$, $\mathbf{x}_A$ denotes the subvector of $\mathbf{x}$ with entries corresponding to indices in A. For any matrix $\mathbf{X} \in \mathbb{R}^{n \times p}$, $\mathbf{X}_A$ denotes the submatrix of $\mathbf{X}$ obtained by selecting the *columns* with indices in A. $S = \{i \in V : \ \beta_i^* \neq 0\}$ denotes the true support of $\boldsymbol{\beta}^*$, its size is

$s = |S|$, and hence $p - s$ is the number of truly inactive parameters. The set of candidate models is denoted by $\mathcal{M}$ and is given by subsets $M \subseteq V$. The set V is partitioned into disjoint blocks $B_j \subseteq V$ for $j = 1, \ldots, b$, where b is fixed. $S_j = \{i \in B_j : \ \beta_i^* \neq 0\}$ denotes the set of active parameters in block B_j, $s_j = |S_j|$ its size, $p_j - s_j$ the number of inactive parameters in the block. We denote by $\beta^*_{\min,j}$ and $\beta^*_{\min}$ lower bounds on the smallest active signal in S_j and S respectively, such that the data-generating $\boldsymbol{\beta}^*$ belongs to the set $\mathfrak{B} = \{\boldsymbol{\beta} \in \mathbb{R}^p : \text{for all } j \in \{1, \ldots, b\} \text{ and } i \in B_j, |\beta_i| \geq \beta^*_{\min,j} \text{ if } i \in S_j \text{ and } \beta_i = 0 \text{ otherwise}\}$.

Given sequences $f(n) > 0$ and $g(n) > 0$, $f(n) = O(g(n))$ means that there exists a constant $c < \infty$ such that $f(n) \leq cg(n)$ for all $n \geq n_0$ and some fixed n_0, $f(n) = o(g(n))$ means that $\lim_{n\to\infty} f(n)/g(n) = 0$, and $f(n) \asymp g(n)$ means that $f(n) = O(g(n))$ and $g(n) = O(f(n))$. We write $f(n) \gtrsim g(n)$ when there exists a constant $c > 1$ and a sequence $d(n) \to \infty$ such that $f(n) \geq cg(n) + d(n)$. For any set A, A^C denotes the complement of A.

2 Externally-informed penalties and transfer learning

We describe externally-informed ℓ_0 penalization and their connection to Bayesian variable selection (Section 2.1), and general assumptions used across our results (Section 2.2).

2.1 Externally-informed penalization

We recall the ℓ_0-penalized approach to variable selection. Consider a set of candidate models $\mathcal{M}$ given by subsets $M \subseteq V$ such that $|M| \leq n$ and their parameter spaces

$$\mathcal{L}_M := \{\beta \in \mathbb{R}^p : \beta_i = 0 \text{ if } i \notin M\}.$$

A standard ℓ_0 procedure with penalty $\eta(M) = \kappa|M|$ for some $\kappa > 0$ selects the model

$$\hat{S}^{sd} = \arg\max_{M \in \mathcal{M}} \left\{ \max_{\beta \in \mathcal{L}_M} \ell(\mathbf{y}; \boldsymbol{\beta}) - \kappa|M| \right\}, \tag{2}$$

where $\ell(\mathbf{y}; \boldsymbol{\beta})$ denotes the log-likelihood function. Both p and the number of models $|\mathcal{M}|$ may far exceed n, and we may only consider models of size $|M| \leq n$ without loss of generality (if $\kappa > 0$ such models are never selected).

A useful way to interpret ℓ_0 penalization is via its connection to Bayesian variable selection under discrete spike-and-slab priors where, for each variable i, one has a binary indicator $m_i = I(\beta_i \neq 0)$ for the inclusion of variable i in the model. Each model $M \in \mathcal{M}$ corresponds then to a particular value of $(m_1, \ldots, m_p)$. The joint prior on parameters and models is

$$p(\boldsymbol{\beta}, M \mid \theta) \;=\; p(\boldsymbol{\beta} \mid M) p(M \mid \theta) \tag{3}$$

where $p(\boldsymbol{\beta} \mid M)$ is a prior on $\boldsymbol{\beta}$ given model M, and variable inclusions are independent a priori with inclusion probability θ, that is

$$p(M \mid \theta) \propto \prod_{i=1}^{p} \mathrm{Bern}\left(m_i; \theta\right). \tag{4}$$

Posterior model probabilities are $p(M \mid \mathbf{y}, \theta) \propto p(\mathbf{y} \mid M) p(M \mid \theta)$, where $p(\mathbf{y} \mid M)$ is the so-called marginal likelihood of model M. The BIC approximation (Schwarz 1978) to $p(M \mid \mathbf{y}, \theta)$ gives that, for a wide family of priors $p(\boldsymbol{\beta} \mid M)$,

$$\ln p(M \mid \mathbf{y}, \theta) \;\approx\; \ln p(\mathbf{y} \mid \tilde{\boldsymbol{\beta}}^{(M)}) - \left(\frac{1}{2}\ln(n) + \ln\left(1/\theta - 1\right)\right)|M| + c_M, \tag{5}$$

where $\tilde{\boldsymbol{\beta}}^{(M)}$ is the maximum likelihood estimator (MLE) under model M, and c_M is a constant that may depend on M. This approximation makes explicit the close relationship between Bayesian model selection and ℓ_0 penalties. Neglecting c_M and retaining only terms that grow with n, simple algebra shows that the ℓ_0 selector in (2) with $\kappa = \frac{1}{2}\ln(n) + \ln(1/\theta - 1)$ approximately maximizes $p(M \mid \mathbf{y}, \theta)$. This gives a correspondence between a choice of prior inclusion probability θ and a choice of ℓ_0 penalty κ.

This Bayesian perspective also suggests how external information can be incorporated into variable selection. Assume that one has external information $\boldsymbol{z} = (z_1, \ldots, z_p)$ on the relevance of variables for association with $\mathbf{y}$. In a spike-and-slab prior, this information can be naturally reflected by setting prior inclusion probabilities that depend on $\boldsymbol{z}$, that is replacing θ in (4) by $\theta(z_i)$, $i = 1, \dots, p$. When $\boldsymbol{z}$ is discrete, it defines blocks of variables that share similar prior relevance. In this work, we assume that $\boldsymbol{z}$ partitions the variables into blocks $B_1, \dots, B_b$, obtaining block-dependent prior probabilities $\theta(z_i) = \theta_j$ for all $i \in B_j$,

$$p(M \mid \boldsymbol{\theta}) \propto \prod_{i=1}^{p} \text{Bern}\,(m_i; \theta_j)\, I(i \in B_j), \tag{6}$$

The BIC approximation for this model prior gives

$$\ln p(M \mid \mathbf{y}, \theta) \;\approx\; \ln p(\mathbf{y} \mid \tilde{\boldsymbol{\beta}}^{(M)}) - \sum_{j=1}^{b} \Big(\frac{1}{2}\ln(n) + \ln\Big(1/\theta_j - 1\Big)\Big)|M_j| + c_M, \tag{7}$$

where $M_j \subseteq B_j$ are the selected variables in block j and $M = \cup_{j=1}^{b} M_j$ the model.

Motivated by this correspondence, we now return to a frequentist perspective. We study in this work the ℓ_0 selector analogue to maximizing $p(M \mid \mathbf{y}, \theta)$ in (7). That is, an informed ℓ_0 selector that penalizes differently the blocks $B_1, \dots, B_b$,

$$\hat{S}^{ei} \;\in\; \arg\max_{M \in \mathcal{M}} \left\{ \max_{\beta \in \mathcal{L}_M} \ell(\mathbf{y}; \boldsymbol{\beta}) - \sum_{j=1}^{b} \kappa_j |M_j| \right\}, \tag{8}$$

where $\kappa_1 > 0, \dots, \kappa_b > 0$. Under the BIC approximation in (7), $\hat{S}^{ei}$ corresponds to the posterior mode with block prior inclusion probabilities $\theta_j = (\exp(\kappa_j - \ln(n)/2) + 1)^{-1}$. In the case that $b = 1$ (all variables form a single block), $\hat{S}^{ei}$ recovers $\hat{S}^{sd}$ in (2). Throughout, the superscripts "sd" and "ei" stand for "standard" and "externally-informed," respectively.

2.2 General assumptions of theoretical results

We study the properties of $\hat{S}^{ei}$ as the sample size $n \to \infty$. Although not explicitly denoted, p, s and $\beta^*_{\min}$ are functions of n, and so are s_j, p_j and $\beta^*_{\min,j}$. In particular the numbers of inactive and active variables ($p_j - s_j$ and s_j) can be either fixed or grow with n. We make no assumption about the regime linking n and p, but the main interest is in $n = o(p)$

settings. For convenience, we assume that in each block j, $p_j - s_j \geq 1$ and $s_j \geq 1$, but our results can easily be adapted to $p_j - s_j = 0$ and $s_j = 0$.

We assume that $\mathcal{M}$ is a well-specified set of candidate models, that is S, the support of $\boldsymbol{\beta}^*$, lies in $\mathcal{M}$, and that one constrains attention to models with full-rank design, such that for any model $M \in \mathcal{M}$, $\text{rank}(\mathbf{X}_M^\top \mathbf{X}_M) = |M|$. This assumption implies $s = |S| \leq n$. We also assume that $\hat{S}^{ei}$ is unique and that the error variance σ^2 is known. To simplify notation, without loss of generality, we set $\sigma^2 = 1$. Non-uniqueness of $\hat{S}^{ei}$, non-full rank models, and unknown σ^2 can be accommodated, at the expense of a slightly more involved treatment.

3 Oracle externally-informed penalties

We discuss the properties of the externally-informed model selector $\hat{S}^{ei}$ in (8) in the linear model (1). Our goal is to show that $P(\hat{S}^{ei} = S) \to 1$ as $n \to \infty$, and that this occurs under milder conditions or at a faster rate than for the standard selector $\hat{S}^{sd}$ in (2), which uses a common penalty across blocks. In Section 3.1, we prove the consistency of $\hat{S}^{ei}$ under sufficient conditions. In Section 3.2 we discuss matching necessary conditions that are important not only to ensure the tightness of our analysis, but also to describe the benefits of $\hat{S}^{ei}$ over $\hat{S}^{sd}$. In Section 3.3, we discuss these benefits in relation to the relevance of the external information, that is to what extent the blocks discriminate between truly active and inactive signals. In this section we assume that penalties are set by an oracle that knows the true pattern of sparsity and signal strength. In Section 3.4, we establish a generalized convergence result for $\hat{S}^{ei}$ under weak conditions on signal strength, which plays an important role for the data-based procedures in Section 4.

3.1 Oracle properties

We first outline our proof strategy. We remark that, although the strategy builds upon that in Rossell (2022), there are significant innovations. Rossell (2022) did not consider block penalties and, when applied to the single penalty case, our sufficient conditions to attain consistency are milder and the associated rates are sharper. We also obtain new necessary conditions (Section 3.2) that nearly match the sufficient conditions. We also compared our results to those obtained under a Gaussian sequence model, a simplified setting where a sharp characterization is possible, obtaining analogous results to our regression setting. Finally, the proof and results in Section 3.4 are new, and these are crucial for the theory of our data-based procedure in Section 4.

For any model $M \in \mathcal{M}$, let $\tilde{\boldsymbol{\beta}}^{(M)} = (\mathbf{X}_M^\top \mathbf{X}_M)^{-1}\mathbf{X}_M^\top \mathbf{y} \in \mathbb{R}^{|M|}$ be the MLE under model M, and

$$C(M) \;:=\; \tfrac{1}{2}\|\mathbf{X}_M \tilde{\boldsymbol{\beta}}^{(M)}\|^2 - \sum_{j=1}^{b} \kappa_j |M_j| \quad \text{and} \quad NC(M) \;:=\; \frac{\exp(C(M))}{\sum\limits_{M' \in \mathcal{M}} \exp(C(M'))}. \tag{9}$$

Lemma 3.1. *$\hat{S}^{ei}$ satisfies $\hat{S}^{ei} \;=\; \arg\max_{M\in\mathcal{M}} NC(M)$*

By Lemma 3.1, $\hat{S}^{ei}$ selects $M \in \mathcal{M}$ with the largest $NC(M)$, which relates to the sum-of-squares explained by M. By the BIC approximation in (7), $C(M)$ and $NC(M)$ can be understood as, respectively, the unnormalized and normalized pseudo-posterior probability for model M in a Bayesian framework. We refer to $C(M)$ and $NC(M)$ as the unnormalized and normalized scores for M, respectively. Lemma 3.2, reproduced from Rossell (2022), proves that the expected sum of the normalized scores for $M \neq S$ bounds the probability of an incorrect model selection.

Lemma 3.2. *(i) $P(\hat{S}^{ei} \neq S) \;\leq\; 2\sum_{M\in\mathcal{M}\setminus\{S\}} \mathbb{E}\left(NC(M)\right)$.*

(ii) For any $M, M' \in \mathcal{M}$, such that $M \neq M'$, $NC(M) \;\leq\; \Big(1 + \exp(C(M') - C(M))\Big)^{-1}$.

Using Lemma 3.2(i), we control $P(\hat{S}^{ei} \neq S)$ by showing that $\sum_{M \neq S} \mathbb{E}\{NC(M)\}$ vanishes asymptotically. That is, we establish strong consistency in the sense that $NC(S) \xrightarrow{L_1} 1$. By Lemma 3.2(ii) with $M' = S$, it suffices to control $\mathbb{E}\{(1+\exp(C(S)-C(M)))^{-1}\}$, which reduces to bounding quadratic forms of least-squares estimators. We obtain these bounds using concentration inequalities and then summing over $M \neq S$.

We now state two assumptions that are sufficient to asymptotically recover S.

(A1) For each j, there exists $f_j \to \infty$ such that,

$$\kappa_j \;=\; \ln(p_j - s_j) + f_j$$

(A2) For each j, there exists $g_j \to \infty$ such that,

$$\sqrt{\frac{(1-\gamma)n\rho(\mathbf{X})}{6}}\beta^*_{\min,j} \;-\; \sqrt{\kappa_j} \;=\; \sqrt{\ln(s_j)} + g_j,$$

where $\rho(\mathbf{X}) = \min_{M \in \mathcal{M}:\, M \not\supseteq S} \lambda_{\min}\Big(n^{-1}\mathbf{X}_{S\setminus M}^\top \left(I_n - \mathbf{X}_M \left(\mathbf{X}_M^\top \mathbf{X}_M\right)^{-1} \mathbf{X}_M^\top\right) \mathbf{X}_{S\setminus M}\Big)$, and $\gamma := \frac{1}{2}(1 + \max_j \ln(p_j - s_j)/\kappa_j) \in (\frac{1}{2}, 1)$.

Assumption A1 sets a lower-bound the block penalties κ_j, whereas Assumption A2 imposes a beta-min condition on the block-wise signal strengths $\beta^*_{\min,j}$. In the simplest orthonormal setting where $\mathbf{X}^\top\mathbf{X} = n\mathbb{I}_p$, the quantity $\rho(\mathbf{X}) = 1$. More generally, $\rho(\mathbf{X}) \geq 0$ measures the correlation between truly active variables $\mathbf{X}_S$ and other variable sets $\mathbf{X}_M$. For further discussion see, e.g., Wainwright (2010). Our proofs allow slightly weaker conditions, but we present A1–A2 in this form to facilitate interpretation and comparison with existing results.

We next state our main theorems on strong variable selection consistency of $\hat{S}^{ei}$ and the associated rates. In Theorem 3.4, the constants $\delta \in (0,1)$ and $\gamma^* \in (1/2, 1)$ should be taken arbitrarily close to 1 and 1/2, respectively. After presenting these results, we examine the tightness of Assumptions A1–A2 relative to necessary conditions for selection consistency.

Theorem 3.3. *Assume A1 and A2, and recall that* $\mathfrak{B} = \{\boldsymbol{\beta} \in \mathbb{R}^p$ *: for all* $j \in$

$\{1,\ldots,b\}$ *and* $i \in B_j$, $|\beta_i| \geq \beta^*_{\min,j}$ *if* $i \in S_j$ *and* $\beta_i = 0$ *otherwise*$\}$. *Then,*

$$\lim_{n\to\infty} \sup_{\boldsymbol{\beta}^*\in\mathfrak{B}} \sum_{M\in\mathcal{M}\setminus\{S\}} \mathbb{E}\left(NC(M)\right) = 0 \qquad \textit{and} \qquad \lim_{n\to\infty} \inf_{\boldsymbol{\beta}^*\in\mathfrak{B}} P(\hat{S}^{ei} = S) = 1.$$

Theorem 3.3 guarantees that the correct selection probability $P(\hat{S}^{ei} = S)$ converges to 1, uniformly across the set of $(s_1,\ldots,s_b)$-sparse $\boldsymbol{\beta}^* \in \mathfrak{B}$.

Theorem 3.4. *Assume A1 and A2. Then, there exists a constant $c > 0$ such that for all sufficiently large n and any $\delta \in (0,1)$,*

$$\sup_{\boldsymbol{\beta}^*\in\mathfrak{B}} P(\hat{S}^{ei} \neq S) \leq c\sum_{j=1}^{b} \exp\left[-\tfrac{\delta}{2}(\kappa_j - \ln(p_j - s_j))\right] + \exp\left[-\tfrac{\delta}{2}\left(\left(\sqrt{\tfrac{(1-\gamma)n\rho(\mathbf{X})\beta^*_{\min,j}}{6}} - \sqrt{\kappa_j}\right)^2 - \ln(s_j)\right)\right]. \tag{10}$$

Moreover, suppose that for some fixed $\gamma^* \in (1/2, 1)$ and all $j = 1,\ldots,b$, the κ_j are set at the oracle values κ^*_j satisfying

$$\sqrt{\kappa^*_j} \;=\; \frac{1}{2}\sqrt{\frac{(1-\gamma^*)n\rho(\mathbf{X})}{6}}\beta^*_{\min,j} + \frac{1}{2}\sqrt{\frac{6}{(1-\gamma^*)n\rho(\mathbf{X})}}\frac{1}{\beta^*_{\min,j}}\Big(\ln(p_j - s_j) - \ln(s_j)\Big), \tag{11}$$

and that $\lim_{n\to\infty} g(\gamma^*)\sqrt{n\rho(\mathbf{X})/6}\;\; \beta^*_{\min,j}/(\sqrt{\ln(p_j - s_j)} + \sqrt{\ln(s_j)}) > 1$ *where* $g(\gamma^*) = \sqrt{(-1+2\gamma^*)(1-\gamma^*)}/(1+\sqrt{2-2\gamma^*})$. *Then Assumptions A1 and A2 hold, and, for all sufficiently large n and any $\delta \in (0,1)$, we have*

$$\sup_{\boldsymbol{\beta}^*\in\mathfrak{B}} P(\hat{S}^{ei} \neq S) \;\leq\; 2c\sum_{j=1}^{b} \exp\Big[-\tfrac{\delta}{2}\Big(\tfrac{(1-\gamma^*)n\rho(\mathbf{X})}{24}{\beta^*_{\min,j}}^2 - \ln\max\{p_j - s_j, s_j\}\Big)\Big]. \tag{12}$$

Theorem 3.4 gives the rate at which $P(\hat{S}^{ei} \neq S)$ vanishes. In (10), each summand features a first term that decays exponentially in the penalty κ_j, and a second term that decays exponentially in $(\sqrt{\frac{(1-\gamma)n\rho(\mathbf{X})}{6}}\,\beta^*_{\min,j} - \sqrt{\kappa_j})^2$. These two terms capture competing type I and II error contributions. The choice $\kappa_j = \kappa^*_j$ balances these contributions by equating their exponential rates, and thus approximately minimizes (10). We refer to κ^*_j as *oracle penalties*, since they depend on the unknown s_j and $\beta^*_{\min,j}$. In (12), we further derive a simplified upper bound for the rate achieved by κ^*_j. This bound is sharpest in regimes where the signal strength term $n\rho(\mathbf{X})\beta^{*2}_{\min,j}$ grows faster than $\ln(p_j/s_j - 1)$.

The results of Theorem 3.4, when applied to uninformed ℓ_0 selector $\hat{S}^{sd}$, nearly match the minimax rates for the orthonormal case, where $\mathbf{X}^\top\mathbf{X} = n\mathbb{I}_p$ and $\rho(\mathbf{X}) = 1$. Butucea et al. (2018) show that, in this case, the choice $\sqrt{\tilde{\kappa}} = (1/2)\sqrt{n/2}\beta^*_{\min} + (1/2)\sqrt{2/n}\ln(p/s-1)/\beta^*_{\min}$ is exactly minimax for the Hamming loss and asymptotically minimax for the probability of incorrect recovery, when $\beta^*_j \geq 0$ for all j. The minimax rate for the Hamming loss is then shown to be the sum of a term decreasing exponentially in $\tilde{\kappa} - \ln(p-s)$ and a term decreasing exponentially in $(\sqrt{n/2}\beta^*_{\min} - \sqrt{\tilde{\kappa}})^2 - \ln(s)$, similarly to our rate in (10).

3.2 Necessary conditions

We compare our sufficient Assumptions A1 and A2 to assumptions on κ_j and $\beta_{\min,j}$ that are necessary for consistent variable selection with $\hat{S}^{ei}$. To obtain tight necessary assumptions, it suffices to require that S is preferred over models that differ by one variable. Consider, for each j, the subset of models that over-fit by only one variable from block B_j,

$$\mathcal{O}_j \;=\; \{M \in \mathcal{M} \mid M = S \cup \{i\} \text{ where } i \in B_j \setminus S_j\}. \tag{13}$$

Our necessary conditions involve the next two quantities

$$\underline{\lambda}_j \;:=\; \lambda^2_{\min}(C^j) \quad\text{ and }\quad \bar{\lambda} \;:=\; \lambda_{\max}\Big(\frac{1}{n}\mathbf{X}_S^\top\mathbf{X}_S\Big), \tag{14}$$

where C^j is the matrix in $\mathbb{R}^{(p_j-s_j)\times(p_j-s_j)}$ such that for any $M^k, M^l \in \mathcal{O}_j$, $C^j_{k,l} = \operatorname{corr}\left(Z_{M^k}, Z_{M^l}\right)$ and $Z^2_{M^k} = \|\mathbf{X}_{M^k}\tilde{\boldsymbol{\beta}}^{(M^k)}\|^2 - \|\mathbf{X}_S\tilde{\boldsymbol{\beta}}^{(S)}\|^2$. That is, $Z^2_{M^k}$ is an increase in explained sum-of-squares measuring the extent to which M^k overfits the observed data relative to S, and hence small $\underline{\lambda}_j$ indicates that truly inactive variables in block j are highly correlated with each other, whereas $\underline{\lambda}_j = 1$ that they are uncorrelated. Similarly, $\bar{\lambda}$ relates to the correlation between active variables.

Proposition 3.5. *(i) If for some $j = 1, \ldots, b$, $p_j - s_j \to \infty$ and $\lim_{n\to\infty}\kappa_j/(\underline{\lambda}_j\ln(p_j - s_j)) < 1$, then $\lim_{n\to\infty}\sup_{\beta^*\in\mathfrak{B}} P(\hat{S}^{ei} = S) = 0$.*

(ii) If for some $j = 1, \ldots, b$, $\kappa_j \to \infty$ and $\lim_{n\to\infty} \sqrt{n\bar{\lambda}}\beta^*_{\min,j} - \sqrt{2\kappa_j} < \infty$ then there exists $\boldsymbol{\beta}^* \in \mathfrak{B}$ such that $\lim_{n\to\infty} P(\hat{S}^{ei} = S) < 1$.

(iii) If for some $j \in \{1, \ldots, b\}$, $\lim_{n\to+\infty}(p_j - s_j)^{\underline{\lambda}_j} = +\infty$ and $\lim_{n\to\infty} \sqrt{n\bar{\lambda}}\beta^*_{\min,j} - \sqrt{2\underline{\lambda}_j \ln(p_j - s_j)} < \infty$, then there exists $\boldsymbol{\beta}^* \in \mathfrak{B}$ such that $\lim_{n\to\infty} P(\hat{S}^{ei} = S) < 1$.

Proposition 3.5 implies that Assumptions A1-A2 are fairly tight. By Part (i), the block penalty κ_j must grow at least as fast as $\ln(p_j - s_j)$. Since $\underline{\lambda}_j \in [0, 1]$, in the worst case where $\underline{\lambda}_j = 1$ Part (i) implies that it is necessary that $\kappa_j \geq \ln(p_j - s_j)$ nearly matching A1. Part (ii) shows that the signal strength $\beta^*_{\min,j}$ must be sufficiently large relative to κ_j, and closely resembles the requirement that $\sqrt{(1-\gamma)n\rho(\mathbf{X})\beta^*_{\min,j}/6} - \sqrt{\kappa_j} - \sqrt{\ln(s_j)} \to \infty$ made by A2, up to logarithmic term in s_j, and to replacing $\rho(\mathbf{X})(1-\gamma)$ by $\bar{\lambda}$, where, recall, $1-\gamma \in (0, 1/2)$ and it is bounded away from 0 if $\ln(p_j - s_j) = O(f_j)$ for all j. Part (iii) follows by combining Parts (i)-(ii), and implies that $\beta^*_{\min,j}$ must satisfy $\lim_{n\to\infty} \sqrt{n\bar{\lambda}}\beta^*_{\min,j} - \sqrt{2\underline{\lambda}_j \ln(p_j - s_j)}$, which again resembles A2 in the worst case where $\underline{\lambda}_j = 1$. That is, Assumption A2 and Parts (ii)-(iii) imply the same scalings of $(n, p_j, s_j, \beta^*_{\min,j})$, up to logarithmic terms.

3.3 Benefits of externally-informed penalties

This section compares the oracle-informed selector $\hat{S}^{ei}$ with the standard ℓ_0 selector $\hat{S}^{sd}$, which possesses the best-known properties in terms of variable selection consistency (Gao & Aragam 2025). We highlight two types of improvements in $\hat{S}^{ei}$: (i) weaker conditions for variable selection consistency and (ii) faster convergence rates when both $\hat{S}^{ei}$ and $\hat{S}^{sd}$ are consistent. Both improvements depend on how the blocks differ in their sparsity levels s_j and signal strengths $\beta^*_{\min,j}$, that is on how informative the blocks truly are.

By Theorem 3.4, Assumptions A1–A2 are sufficient for consistent support recovery, both for $\hat{S}^{ei}$ and $\hat{S}^{sd}$ (as the particular case where $b = 1$). To compare $\hat{S}^{ei}$ and $\hat{S}^{sd}$, we plug in the smallest penalties κ_j allowed by A1–A2 and inspect the implied scaling of the sample size n.

Consider a high-dimensional setting where for all j, $p_j - s_j \to \infty$. For the informed selector $\hat{S}^{ei}$, with block penalties $\kappa_j \asymp \ln(p_j - s_j)$, consistency requires that

$$\sqrt{n} \gtrsim \frac{\sqrt{\ln(p_j - s_j)} + \sqrt{\ln(s_j)}}{\sqrt{\rho(\mathbf{X})}\,\beta^*_{\min,j}} \quad \text{for each } j = 1, \ldots, b. \tag{15}$$

For the standard $\hat{S}^{sd}$, with a single penalty $\kappa \asymp \ln(p - s)$, consistency requires

$$\sqrt{n} \gtrsim \frac{\sqrt{\ln(p - s)} + \sqrt{\ln(s)}}{\sqrt{\rho(\mathbf{X})}\,\beta^*_{\min}}. \tag{16}$$

Since $p_j - s_j \leq p - s$, $s_j \leq s$, and $\beta^*_{\min,j} \geq \beta^*_{\min}$ for all j, (15) is always weaker than (16).

The magnitude of this improvement depends on how strongly $(\beta^*_{\min,j}, s_j, p_j - s_j)$ vary across blocks, whether the blocks truly capture variables with different signal strengths or sparsity levels. To illustrate, we discuss next the requirement on signals strength implied by (15) and (16). That is, for the informed selector $\hat{S}^{ei}$,

$$\beta^*_{\min,j} \gtrsim \frac{\sqrt{\ln(p_j - s_j)} + \sqrt{\ln(s_j)}}{\sqrt{\rho(\mathbf{X})n}} \quad \text{for each } j = 1, \ldots, b, \tag{17}$$

and for the standard $\hat{S}^{sd}$,

$$\beta^*_{\min} \gtrsim \frac{\sqrt{\ln(p - s)} + \sqrt{\ln(s)}}{\sqrt{\rho(\mathbf{X})n}}. \tag{18}$$

The right-hand sides in (17) and (18) are smallest signal strength allowed by the sufficient assumptions for consistency (A1–A2) with $\hat{S}^{ei}$ and $\hat{S}^{sd}$ respectively. Consider a $b = 2$ blocks example with $p - s = n^2$ inactive variables: $p_1 - s_1 = n^2 - n^{1/2}$ in block 1 and $p_2 - s_2 = n^{1/2}$ in block 2. Suppose also that $s_1 = s_2 = O(1)$ and that the design is orthonormal, so that $\rho(\mathbf{X}) = 1$. For the standard selector $\hat{S}^{sd}$, consistency requires $\beta^*_{\min} \gtrsim \sqrt{2 \ln n/n}$. In contrast, for the informed selector $\hat{S}^{ei}$, consistency requires $\beta^*_{\min,1} \gtrsim \sqrt{2 \ln n/n}$ in block 1, but only $\beta^*_{\min,2} \gtrsim \sqrt{\ln n/(2n)}$ in block 2. That is, $\hat{S}^{ei}$ can recover signals twice smaller than $\hat{S}^{sd}$ in block 2. Additional scenarios illustrating how gains depend on block sparsity and signal heterogeneity across blocks are provided in Supplement S1.

The comparison above is based on sufficient conditions. Proposition 3.5(iii) shows that a similar comparison holds for necessary conditions. In particular, there exist regimes in which the sample size n grows insufficiently quickly for consistent recovery using $\hat{S}^{sd}$, but sufficiently for $\hat{S}^{ei}$. Corollary 3.6 formalizes this result. Corollary 3.6 assumes a high-dimensional setting, where $\underline{\lambda}\ln(p-s)$ diverges, as in Proposition 3.5 (iii).

Corollary 3.6. *Assume A1, A2 and* $\lim_{n\to\infty}(p-s)^{\underline{\lambda}} = \infty$ *for* $\underline{\lambda}$ *as in* (14) *for the* $b=1$ *block case. If* $\lim_{n\to\infty}\sqrt{n\bar{\lambda}}\beta^*_{\min} - \sqrt{2\underline{\lambda}\ln(p-s)} < \infty$*, then there exists* $\boldsymbol{\beta}^* \in \mathfrak{B}$ *such that* $\lim_{n\to\infty} P(\hat{S}^{sd} = S) < 1$ *and* $\lim_{n\to\infty}\inf_{\boldsymbol{\beta}^*\in\mathfrak{B}} P(\hat{S}^{ei} = S) = 1$.

Supplement S1 provides an illustration (simulation scenario 2).

We next compare the probabilities of correct selection when both procedures are consistent. Assume that block penalties are set to their oracle values κ_j^* in Theorem 3.4, and that the standard penalty is set to κ^*. Let OR^{ei} and OR^{sd} denote the corresponding oracle convergence rates in (12). Up to constant factors, their ratio satisfies

$$\frac{OR^{ei}}{OR^{sd}} = \sum_{j=1}^{b} \exp\left(-\frac{\delta}{2}\left[\frac{n\rho(\mathbf{X})(1-\gamma^*)}{24}\left({\beta^*_{\min,j}}^2 - {\beta^*_{\min}}^2\right) + \ln\frac{\max\{p-s,s\}}{\max\{p_j-s_j,s_j\}}\right]\right), \tag{19}$$

where δ can be taken arbitrarily close to 1. Since $\beta^*_{\min,j} \geq \beta^*_{\min}$ and $\max\{p_j-s_j,s_j\} \leq \max\{p-s,s\}$, it follows that $OR^{ei} = O(OR^{sd})$ for any block partition. Moreover, if for every j, $\beta^*_{\min,j} > \beta^*_{\min}$ or $(p-s)/(p_j-s_j) \to \infty$, then $OR^{ei} = o(OR^{sd})$, implying faster convergence. Consider a sparse setting where $s_j < p_j - s_j$ for all j. There are then two sources of improvement in convergence rates: in any block j, a smaller number of truly zero parameters, that is $p_j - s_j < p - s$, or a larger signal, $\beta^*_{\min,j} > \beta^*_{\min}$. In blocks where $\beta^*_{\min,j} \approx \beta^*_{\min}$, improvement comes mainly from having a smaller number of truly zero means. In those blocks, the oracle sets $\kappa_j^* < \kappa^*$, so that smaller signals are detected. If $\beta^*_{\min,j}$ is much greater than $\beta^*_{\min}$, the oracle sets $\kappa_j^* > \kappa^*$ and the probability of false positives is reduced. Further illustrations are provided in Supplement S1.

3.4 Generalized convergence

We now discuss a convergence result for $\hat{S}^{ei}$ that is a key ingredient for the data-driven procedures in Section 4. This result generalizes Theorem 3.3 by dropping the beta-min Assumption A2. Instead, active signals are separated into three categories—*small*, *intermediate*, and *large*—relative to the penalties κ_j. We define next these three categories.

For fixed block penalties $\kappa_1, \ldots, \kappa_b$, and for γ and g_j as in Assumption A2, define

$$\begin{aligned} S_j^L(\kappa_j) \ &:= \ \left\{ i \in S_j \,\middle|\, \sqrt{\frac{(1-\gamma)n\rho(\mathbf{X})}{6}}\,|\beta_i^*| - \sqrt{\kappa_j} \ \geq \ \sqrt{\ln(s_j)} + g_j \right\}, \\ S_j^S(\kappa_j) \ &:= \ \left\{ i \in S_j \,\middle|\, \sqrt{n\bar{\lambda}}\,|\beta_i^*| = o\Big(\sqrt{\kappa_j}\Big) \right\}, \\ S_j^I(\kappa_j) \ &:= \ S_j \setminus \Big(S_j^L(\kappa_j) \cup S_j^S(\kappa_j)\Big). \end{aligned} \tag{20}$$

where $\bar{\lambda} = \lambda_{\max}\Big(\mathbf{X}_S^\top \mathbf{X}_S / n\Big)$. The set $S_j^L(\kappa_j)$ contains the large parameters, in that they meet the beta-min Assumption A2. The set $S_j^S(\kappa_j)$ contains small signals relative to the penalty κ_j. The remaining active variables form the intermediate set $S_j^I(\kappa_j)$.

Let $\boldsymbol{\kappa} = (\kappa_1, \ldots, \kappa_b)$ and define $S^L(\boldsymbol{\kappa}) = \cup_{j=1}^b S_j^L(\kappa_j)$, $S^S(\boldsymbol{\kappa}) = \cup_{j=1}^b S_j^S(\kappa_j)$, and $S^I(\boldsymbol{\kappa}) = \cup_{j=1}^b S_j^I(\kappa_j)$. Theorem 3.7 shows that, under Assumption A1, the normalized scores $NC(M)$ in (9) concentrate on models that include all large signals, exclude all inactive variables and all small signals, and differ only in which intermediate variables they include:

$$\mathcal{T}(\boldsymbol{\kappa}) \ := \ \left\{ M \in \mathcal{M} \,\middle|\, S^L(\boldsymbol{\kappa}) \subseteq M \subseteq S^L(\boldsymbol{\kappa}) \cup S^I(\boldsymbol{\kappa}) \right\}. \tag{21}$$

We require the following assumption.

(A3) For each block j, $|S_j^I(\kappa_j)| = O(1)$ and $|S_j^S(\kappa_j)| = O(p_j - s_j)$.

Assumption A3 is mild and can be relaxed: $|S_j^I(\kappa_j)|$ can grow moderately with $p_j - s_j$ in our proof, and one expects $|S_j^I(\kappa_j)|$ to be small because there is a relatively small gap between the ranges of β_i^*'s included in S_j^S and S_j^L. Further, if $|S_j^S(\kappa_j)|/(p_j - s_j) \to \infty$ then Theorem 3.7 still holds, replacing, $p_j - s_j$ by $|S_j^S(\kappa_j)|$ in A1.

Theorem 3.7. *Assume A1 and A3, for every $j = 1, \ldots, b$, then,*

$\lim_{n\to\infty} \sum_{M\in\mathcal{M}\setminus\mathcal{T}(\boldsymbol{\kappa})} \sup_{\boldsymbol{\beta}^*\in\mathfrak{B}} \mathbb{E}\left(NC(M)\right) = 0$ *and* $\lim_{n\to\infty} \inf_{\boldsymbol{\beta}^*\in\mathfrak{B}} P(\hat{S}^{ei} \in \mathcal{T}(\boldsymbol{\kappa})) = 1.$

4 Data-based externally-informed penalties

We now propose a data-driven method to set the block penalties $\kappa_1, \ldots, \kappa_b$ without relying on an oracle. The main idea is to adapt $\kappa_1, \ldots, \kappa_b$ to the sparsity in each block. Our construction uses the Bayesian interpretation of ℓ_0 penalties via the BIC approximation in (7). This link suggests an empirical Bayes strategy for estimating, in each block B_j, the number active coefficients s_j. Although the motivation is Bayesian, the procedure is fully data-based and does not require specifying a prior on $(\boldsymbol{\beta}, M)$.

Recall that under the BIC approximation in (7), choosing κ_j is equivalent to choosing a block-wise prior inclusion probability $\theta_j = (\exp(\kappa_j - \ln(n)/2) + 1)^{-1}$, which is a prior guess for s_j/p_j, the proportion of active coefficients in block B_j. A standard empirical Bayes strategy to set $\hat{\theta}_j$ is to maximize the marginal likelihood of $\mathbf{y}$ given $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_b)$, i.e.

$$\hat{\boldsymbol{\theta}} := \arg\max_{\boldsymbol{\theta}} p(\mathbf{y} \mid \boldsymbol{\theta}) = \arg\max_{\boldsymbol{\theta}} \int p(\mathbf{y} \mid \boldsymbol{\beta}, M) dP(\boldsymbol{\beta}, M \mid \boldsymbol{\theta}).$$

Lemma 4.1 shows that $\hat{\boldsymbol{\theta}}$ satisfies a fixed point equation, whereby the prior inclusion probability $\hat{\theta}_j$ for block j equals the average posterior inclusion probability across variables in that block given $\hat{\boldsymbol{\theta}}$.

Lemma 4.1. *Let $p(M \mid \boldsymbol{\theta}) = \prod_{i=1}^{p} \text{Bern}\left(m_i; \theta_j\right) I(i \in B_j)$ as in (6) and $P(\beta_j \neq 0 \mid \mathbf{y}, \boldsymbol{\theta}) = \sum_{M\in\mathcal{M}: j\in M} P(M \mid \mathbf{y}, \boldsymbol{\theta})$ be the posterior inclusion probability of variable j. Then $\hat{\boldsymbol{\theta}}$ satisfies*

$$\hat{\theta}_j = \frac{1}{p_j} \sum_{i\in B_j} P(\beta_i \neq 0 \mid \mathbf{y}, \hat{\boldsymbol{\theta}}).$$

Motivated by Lemma 4.1 we estimate s_j, the number of active variables in block B_j by the sum of pseudo-posterior inclusion probabilities, computed using $NC(M)$ in (9) as a proxy for $P(M \mid \mathbf{y}, \boldsymbol{\theta})$:

$$\hat{s}_j := \sum_{i \in B_j} \sum_{M \in \mathcal{M}: i \in M} NC(M). \tag{22}$$

We have the following asymptotic bounds on the expectation of $\hat{s}_j/p_j$.

Proposition 4.2. *Assume A1 and A3. Then*

$$\frac{|S_j^L(\kappa_j)|}{p_j} \leq \lim_{n \to \infty} \mathbb{E}\left(\frac{\hat{s}_j}{p_j}\right) \leq \frac{s_j - |S_j^S(\kappa_j)|}{p_j} \quad \textit{for all } j = 1, \ldots, b \textit{ and } \boldsymbol{\beta}^* \in \mathfrak{B}.$$

As a consequence, if the betamin condition in Assumption A2 also holds, then $\hat{s}_j/p_j$ is an asymptotically unbiased estimator of s_j/p_j because $S_j = S_j^L(\kappa_j)$ and $S_j^S(\kappa_j) = \emptyset$ for all j. The proposition also shows that, when Assumption A2 is not met, $\hat{s}_j/p_j$ is asymptotically downward biased by at least the proportion of small signals $|S_j^S(\kappa_j)|/p_j$, but it is guaranteed to be larger than the proportion of large signals $|S_j^L(\kappa)|/p_j$ (i.e., those satisfying A2).

We propose a two-step procedure that uses $\hat{s}_j$ in (22) to set adaptive block penalties that leverage the difference in sparsity across blocks. In the first step, we use a single common penalty across blocks κ° and obtain the corresponding $\hat{s}_j$. In the second step, we use $\hat{s}_j$ to set block penalties $\kappa_j^A = \ln(p_j - \hat{s}_j) + \frac{1}{2}\ln(n)$. This choice of penalties amounts to setting $\theta_j = (p_j - \hat{s}_j + 1)^{-1}$ in the BIC approximation (7). We briefly discuss alternative choices of θ_j at the end of the section.

Algorithm 1 - Adaptive informed selector $\hat{S}^{A,ei}$

1. Set $\kappa_j = \kappa^\circ = \ln(p) + \frac{1}{2}\ln(n)$ for $j = 1, \ldots, b$. Compute $\hat{s}_j/p_j$ in (22) for $j = 1, \ldots, b$.
2. Obtain $\hat{S}^{A,ei}$ solving (8) with $\kappa_j^A = \ln(p_j - \hat{s}_j) + \frac{1}{2}\ln(n)$.

We refer to $\hat{S}^{A,ei}$ as the adaptive informed selector. Step 1 can be carried out with any Bayesian computational method that returns posterior inclusion probabilities $P(\beta_i \neq 0 \mid \mathbf{y}, \boldsymbol{\theta})$. Step 2 can be done with any computational method to find the model with best informed ℓ_0 criterion. In our examples in Section 6 we use MCMC for both steps.

In the oracle analysis of Section 3, the selection properties depend on the number of inactive variables $p_j - s_j$ in each block. In practice, in Algorithm 1, Step 1 may fail to detect small signals, so $p_j - \hat{s}_j$ captures the number of variables that remain effectively indistinguishable from zero in Step 1. That is, $p_j - |S_j^L(\kappa^\circ)|$ where $S_j^L(\kappa^\circ)$ is the set of large signals as defined in (20), evaluated at the Step 1 penalty κ°. This leads to a beta-min condition stated in terms of $p_j - |S_j^L(\kappa^\circ)|$ for $j = 1, \ldots, b$,

(A4) For each block j, there exists $d_j \to \infty$, such that

$$\sqrt{\frac{(1-\xi)n\rho(\mathbf{X})}{2}}\beta^*_{\min,j} - \sqrt{\ln\left(p_j - |S_j^L(\kappa^\circ)|\right) + \frac{1}{2}\ln(n)} = \sqrt{\ln(s_j)} + d_j,$$

where $\xi = \frac{1}{2}(1 + \max_j \ln(p_j - s_j)/(\ln(p_j - s_j) + \frac{1}{2}\ln(n)))$.

Under Assumptions A3 and A4, $\hat{S}^{A,ei}$ consistently recovers the true support S.

Theorem 4.3. *Assume that κ° satisfies A3 and that A4 holds. Then* $\lim_{n\to\infty} \inf_{\beta^*\in\mathfrak{B}} P(\hat{S}^{A,ei} = S) = 1$.

We compare next $\hat{S}^{A,ei}$ to its uninformed counterpart $\hat{S}^{A,sd}$ that sets in Step 2 a single common penalty $\kappa^A = \ln(p - \hat{s}) + \frac{1}{2}\ln(n)$. By Theorem 4.3, $\hat{S}^{A,sd}$ is variable selection consistent under the betamin assumption

$$\sqrt{\frac{(1-\xi')n\rho(\mathbf{X})}{2}}\beta^*_{\min} - \sqrt{\ln\left(p - |S^L(\kappa^\circ)|\right) + \frac{\ln(n)}{2}} = \sqrt{\ln(s)} + d, \tag{23}$$

where $\xi' = \frac{1}{2}(1 + \ln(p - s)/(\ln(p - s) + \frac{1}{2}\ln(n)))$ and $d \to \infty$. Since $\ln(p - |S^L(\kappa^\circ)|) \geq \ln(p_j - |S_j^L(\kappa^\circ)|)$ and $\ln(s) \geq \ln(s_j)$, Assumption A4 for $\hat{S}^{A,ei}$ is milder than (23) for $\hat{S}^{A,sd}$, in particular in less sparse blocks where $p_j - |S_j^L(\kappa^\circ)|$ is smaller than $p - |S^L(\kappa^\circ)|$. Although not shown here, necessary betamin assumptions are also milder for $\hat{S}^{A,ei}$ than for $\hat{S}^{A,sd}$.

A natural alternative is to set prior inclusion probabilities to $\theta_j = \hat{s}_j/p_j$, the empirical Bayes estimate of the proportion of active parameters in block j, which leads to setting $\kappa_j^{EB} = \ln(p/\hat{s}_j - 1)$ in Step 2 of Algorithm 1. The consistency of such a procedure can be proven under a mild additional assumption that bounds the number of active parameters. Compared to the uninformed counterpart with common penalty $\kappa^{EB} = \ln(p/\hat{s} - 1)$, the resulting informed selector requires milder conditions for consistency in denser blocks where $\hat{s}_j/p_j > \hat{s}/p$, and harsher conditions in the sparser blocks where $\hat{s}_j/p_j < \hat{s}/p$. For example, if the global $\beta^*_{\min}$ is located in a sparse block (small $\hat{s}_j/p_j$), then this alternative procedure results in low power relative to Algorithm 1.

5 Structural transfer learning

We now study a structural transfer learning method built on the adaptive informed selector $\hat{S}^{A,ei}$. We assume that for each source dataset $k = 1, \dots, K$ we are given a set of selected variables $\hat{S}^{(k)}$, and we want to use these sets to improve variable selection in a target dataset. The hope is that the sources and target are similar regarding the *support* (truly active variables), in contrast to transfer learning for estimation and prediction (Li et al. 2021, Tian & Feng 2023, Abba et al. 2024) where similarity is expected in the parameter values.

A naive strategy would force the inclusion of any variable selected in at least one source dataset. This can lead to *negative transfer*: if a variable is truly active in a source dataset but not in the target dataset, one commits a type I error. In the worst case, the set of truly active variables in the source and target may not overlap at all. Instead, we use the source selections $\hat{S}^{(k)}$ to define external information $\boldsymbol{z}$ that partitions variables into blocks. In a first step, for each variable $i = 1, \dots, p$, we define $z_i = I(i \in \cup_{k=1}^{K} \hat{S}^{(k)})$. That is, variables in the target dataset are split into a first block containing those that were selected in ≥ 1 source datasets, and a second block gathering the rest. In a second step, we use the adaptive informed selector $\hat{S}^{A,ei}$ with that partition to select variables in the target dataset. This procedure provably prevents negative transfer (asymptotically).

Algorithm 2 - Tran-s-ell0 (a simple two-block transfer rule)

(i) Construct $B_1 = \cup_{k=1}^K \hat{S}^{(k)}$ and $B_2 = V \setminus B_1$.

(ii) Run Algorithm 1 on the target dataset using the block partition (B_1, B_2), and denote the output by $\hat{S}^{\mathrm{tr}}$.

Alternative strategies that construct more than two blocks from the source selections are possible, e.g., by grouping variables by how many source datasets selected them. Our theory continues to apply once a block partition is fixed. More flexible approaches regress inclusion probabilities on the full vector $(I\{i \in \hat{S}^{(1)}\}, \ldots, I\{i \in \hat{S}^{(K)}\})$ (van de Wiel et al. 2019). Such approaches require a separate theoretical treatment and are left for future research.

The theoretical properties of our algorithm follow directly from Theorem 4.3, applied to the blocks defined using the $\hat{S}^{(k)}$'s. Let

$$\begin{aligned}
s_1 &= |B_1 \cap S| = |\cup_{k=1}^K \hat{S}^{(k)} \cap S|, \\
s_2 &= |B_2 \cap S| = |S \setminus \cup_{k=1}^K \hat{S}^{(k)}|, \\
p_1 - s_1 &= |B_1 \setminus S| = |(\cup_{k=1}^K \hat{S}^{(k)}) \setminus S|, \\
p_2 - s_2 &= |B_2 \setminus S| = |V \setminus (\cup_{k=1}^K \hat{S}^{(k)} \cup S)|, \\
p_1 - |S_1^L(\kappa^\circ)| &= |B_1 \setminus S^L(\kappa^\circ)| = |(\cup_{k=1}^K \hat{S}^{(k)}) \setminus S^L(\kappa^\circ)|, \\
p_2 - |S_2^L(\kappa^\circ)| &= |B_2 \setminus S^L(\kappa^\circ)| = |V \setminus (\cup_{k=1}^K \hat{S}^{(k)} \cup S^L(\kappa^\circ))|,
\end{aligned} \tag{24}$$

where $S^L(\kappa^\circ)$ is the subset of large active signals in S as defined in (20) for κ° as in Step 1 of Algorithm 1.

Consider the betamin assumption

(A5) There exist $d_1, d_2 \to \infty$ such that,

$$\sqrt{\frac{(1-\xi)n\rho(\mathbf{X})}{6}}\beta^*_{\min,1} - \sqrt{\ln\left(|(\cup_{k=1}^K \hat{S}^{(k)}) \setminus S^L(\kappa^\circ)|\right) + \frac{\ln(n)}{2}} = \sqrt{\ln\left(|\cup_k S^{(k)} \cap S|\right)} + d_1,$$

and,

$$\sqrt{\frac{(1-\xi)n\rho(\mathbf{X})}{6}}\beta^*_{\min,2} - \sqrt{\ln\left(|V \setminus (\cup_{k=1}^K \hat{S}^{(k)} \cup S^L(\kappa^\circ))|\right) + \frac{\ln(n)}{2}} = \sqrt{\ln\left(|S \setminus \cup_{k=1}^K \hat{S}^{(k)}|\right)} + d_2,$$

where $\xi = \frac{1}{2}(1 + \max_j \ln(p_j - s_j)/(\ln(p_j - s_j) + \frac{1}{2}\ln(n)))$. Assumption A5 is Assumption A4, plugging in the number of active (s_j) and inactive ($p_j - s_j$) parameters in (24).

Corollary 5.1. *Assume that κ° satisfies A3 and that A5 holds. Then $\lim_{n\to\infty} \inf_{\beta^* \in \mathfrak{B}} P(\hat{S}^{tr} = S) = 1$.*

Corollary 5.1 shows that Tran-s-ell0 is selection consistent under milder conditions than a standard, uninformed, ℓ_0 selector $\hat{S}^{A,sd}$, which, recall, is consistent under (23). The larger the overlap between the set of truly active variables S in the target dataset and those selected in the source datasets, $\cup_{k=1}^K \hat{S}^{(k)}$, the milder A5 becomes relative to (23). Corollary 5.1 continues to apply even when $\cup_{k=1}^K \hat{S}^{(k)}$ provides a poor guess for S, making Tran-s-ell0 robust to negative transfer. In that case however, Trans-s-ell0 provides no advantage over standard ℓ_0 criteria.

Remarkably, Tran-s-ell0 achieves gains under weaker requirements than transfer learning algorithms for parameter estimation and prediction. The latter either assume that the sets of variables are the same between source and target datasets (Abba et al. 2024), or make strong assumptions on the similarity of design matrices (Li et al. 2021, Tian & Feng 2023). This is required to avoid negative transfer because parameter estimates are highly sensitive to covariate correlation structures. The gains in estimation that are achievable through transfer learning were in fact shown to be inherently limited by the dissimilarity between source and target design matrices (Liu et al. 2025), an issue also known as the *covariate shift* problem. Our results on structural transfer learning require no assumption on the similarity of design matrices: the variable sets in the source and target datasets may differ largely, even in size.

In addition, many (but not all) transfer learning algorithms for parameter estimation and prediction, e.g. Jiang et al. (2016) and Li et al. (2021), require access to the source datasets. Tran-s-ell0 only requires sets of selected variables as input from the source. It is then more widely applicable and enjoys stronger privacy and communication efficiency guarantees.

6 Numerical illustrations

We illustrate the performance of the adaptive informed selector $\hat{S}^{A,ei}$ (Algorithm 1) on simulated data under different levels of relevance of the external information (Section 6.1). We then illustrate the performance of Tran-s-ell0 (Algorithm 2) on simulated and empirical data (Section 6.2). In Step 1 of Algorithm 1 (and hence also within Algorithm 2), we use $\kappa^\circ = \frac{1}{2}\ln(n) + \ln(p)$ and the MCMC algorithm in function `bestIC` in R package `mombf`. Briefly, each model is assigned a pseudo-posterior probability given by the BIC approximation (5), and an MCMC is used to sample from this distribution. In Step 2 of Algorithm 1, we obtain $\hat{S}^{A,ei}$ by re-scoring the models visited by the MCMC in Step 1 and selecting the best-scoring one. We obtain similarly, for comparison purposes, $\hat{S}^{A,sd}$, the uninformed counterpart of $\hat{S}^{A,ei}$ (using a single common penalty). We also compare $\hat{S}^{A,ei}$ and Tran-s-ell0 to the standard ℓ_0 penalty EBIC (Chen & Chen 2008), the LASSO (Tibshirani 1996) and SCAD (Fan & Li 2001). For the last two methods, the penalization parameter is chosen by cross-validation.

6.1 Adaptive informed selector

We simulate data from the linear model (1) with standard Gaussian errors ($\sigma^2 = 1$), sample size $n \in [20, 700]$, and variable values drawn from a multivariate Gaussian with zero means, unit variances, and pairwise correlations $\mathrm{cor}(x_{ij}, x_{il}) = 0.5$ for $j \neq l$.

Table 1 summarizes our three simulation scenarios, which differ in the sparsity and the informativeness of the blocks. In Scenarios 1 and 2, the blocks are informative: block B_1 is sparser than B_2. In Scenario 1, the blocks are moderately informative, in that $[s_2/(p_2 - s_2)]/[s_1/(p_1 - s_1)] = O(\sqrt{n})$. In Scenario 2, the blocks are highly informative: $\boldsymbol{\beta}^*$

Table 1: Sparsity and block assumptions in simulation scenarios 1–3

Scenario	$p-s$	s	$s_1/(p_1-s_1)$	$s_2/(p_2-s_2)$	$\beta^*_{\min,1}$	$\beta^*_{\min,2}=\beta^*_{\min}$
1	$3n/2$	$3\ln(n)$	$3\ln(n)/(3n-2\sqrt{n})$	$3\ln(n)/(2\sqrt{n})$	0.5	0.33
2	n	$3\ln(n)$	$3\ln(n)/(2n-2\ln(n))$	$3/2$	0.5	0.33
3	n	$3\ln(n)$	$3\ln(n)/n$	$3\ln(n)/n$	0.5	0.5

is sparse overall, but it is non-sparse within block 2 ($s_2 > p_2 - s_2$). In Scenario 3, the blocks are non-informative: each has half of the inactive parameters. In all scenarios, the nonzero β_i^*, $i \in S$, are set uniformly spaced in $(1/2, 1)$ except for the smallest parameter in each block, which takes the values $\beta^*_{\min,1}$ and $\beta^*_{\min,2}$ specified in the last two columns of Table 1.

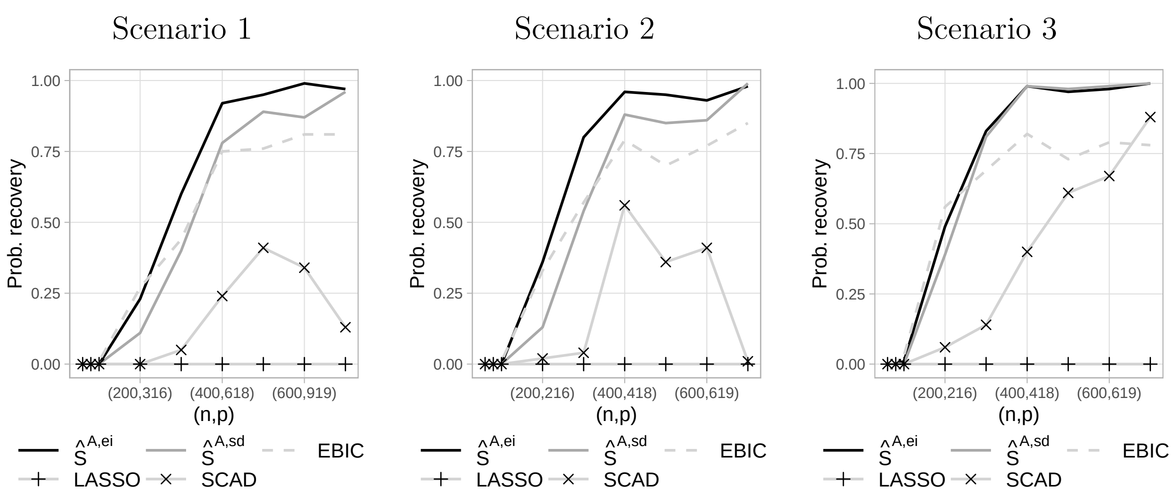

Figure 1: Probability of correct selection with externally-informed $\hat{S}^{ei}$, non-informed $\hat{S}^{A,sd}$, EBIC, LASSO and SCAD in Scenario 1 (left), 2 (middle), 3 (right).

For each sample size, we generate 100 datasets and report in Figure 1 the empirical probability of correct recovery for $\hat{S}^{A,ei}$, $\hat{S}^{A,sd}$, EBIC, LASSO and SCAD. In Scenario 1 and 2 where blocks are discriminative, our externally-informed $\hat{S}^{A,ei}$ outperforms the non-informed $\hat{S}^{A,sd}$, and the EBIC. In Scenario 3, where blocks are non-discriminative, $\hat{S}^{A,ei}$ performs similarly to $\hat{S}^{A,sd}$. LASSO and SCAD have significantly lower probability of

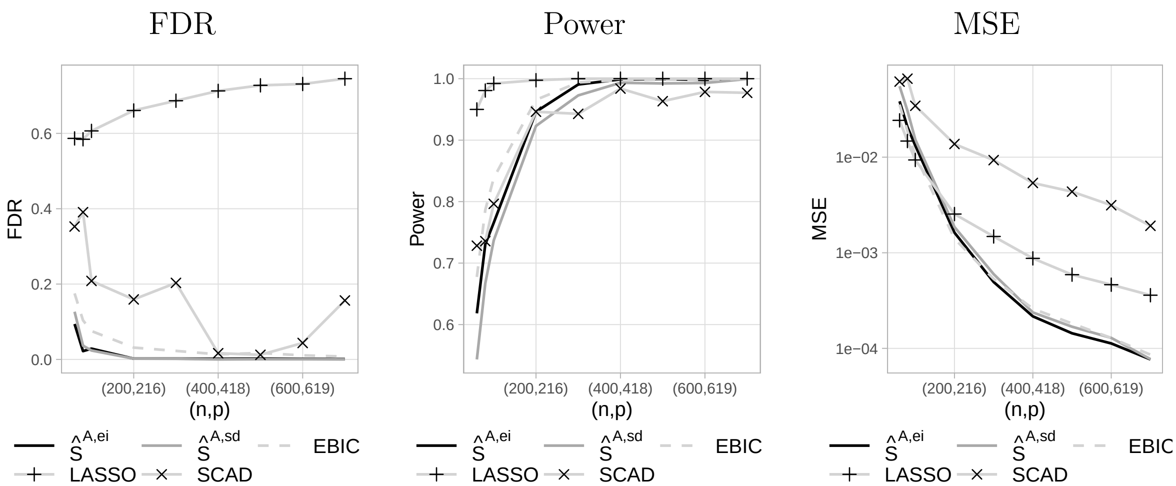

Figure 2: Simulation scenario 2. FDR (left), power (middle) and estimation MSE (right) with $\hat{S}^{A,ei}$, $\hat{S}^{A,sd}$, EBIC, LASSO and SCAD

recovery than the ℓ_0 methods. The average false discovery rate (FDR) and power in Scenario 2, reported in Figure 2, shed light on these differences. Compared to the uninformed $\hat{S}^{A,sd}$, our informed $\hat{S}^{A,ei}$ has equally low or lower FDR and higher power. LASSO and EBIC have higher power than $\hat{S}^{A,ei}$ but higher FDR. In terms of mean squared error, $\hat{S}^{A,ei}$, $\hat{S}^{A,sd}$ and EBIC perform similarly, and better than SCAD and LASSO. Similar findings are observed in Scenarios 1 and 3 (Figure S3 in Supplement S2). We also report in Figure S4 in Supplement S2 the average computation time to obtain $\hat{S}^{A,sd}$ as a function of dimension p. In all three settings, the average time was below 20 seconds for all (n, p).

6.2 Structural transfer learning with Tran-s-ell0

We apply next the Tran-s-ell0 algorithm to simulated data. We consider target dataset sample sizes $n \in [20, 700]$ and, for each sample size, obtain 100 simulations. In each target dataset, the data-generating $\boldsymbol{\beta}^*$ has $p - s = 3n/2$ zero entries and $s = 3\ln(n)$ non-zero entries uniformly spaced in $(1/3, 1)$. For each target dataset, we generate two source datasets with sample size 700 and data-generating parameters that are equal to the target $\boldsymbol{\beta}^*$, except

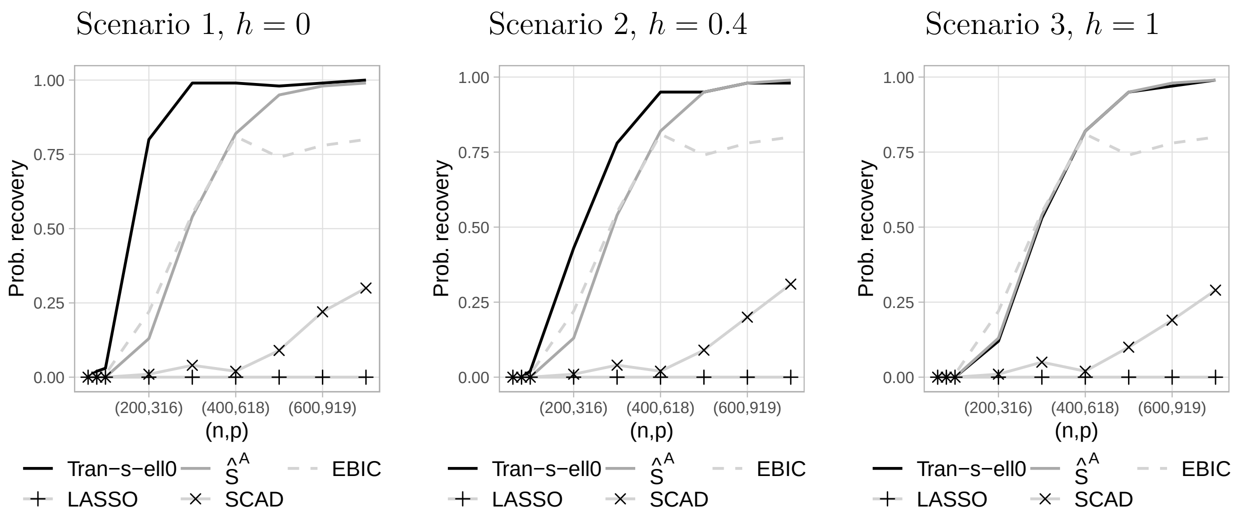

Figure 3: Probability of correct selection with Tran-s-ell0 algorithm, $\hat{S}^{A,sd}$, EBIC, LASSO and SCAD in Scenario 1 (left), 2 (middle), 3 (right).

that a proportion h of the nonzero β_i^*'s are set to 0 and an equal number of the zero β_i^*'s are set to be uniformly spaced in $(1/3, 1)$. We consider three scenarios. In Scenario 1, the truly active variables in the source and target sets are the same, that is $h = 0$. In Scenario 2, there is a moderate overlap, with $h = 0.4$. Scenario 3 is an example of completely irrelevant transfer where $h = 1$. In the target datasets, variable values are drawn from a multivariate Gaussian with zero means, unit variances and pairwise correlations $\text{cor}(x_{ij}, x_{il}) = 0.5$. In the source datasets, variable values are drawn from a multivariate t-distribution with 8 degrees of freedom and pairwise correlations $\text{cor}(x_{ij}, x_{il}) = 0.5^{|i-j|}$. We select variables in the source datasets with a standard ℓ_0 penalty $\kappa = \frac{1}{2}\ln(n) + \ln(p)$ and feed the selected sets to Tran-s-ell0.

Figure 3 shows the average probability of recovery. In Scenarios 1 and 2, Tran-s-ell0 algorithm outperforms all other methods. In Scenario 3, Tran-s-ell0 performs similarly to $\hat{S}^{A,sd}$, better than LASSO and SCAD, slightly worse than EBIC for small n but significantly better for large n. These results support that Tran-s-ell0 is robust to negative transfer.

6.3 Colon cancer data

In molecular biology, it is common to use animal models (e.g. mice) to study genomic mechanisms. Findings from such studies often serve as a basis to study human diseases, for example, to assess whether genes that played key roles in the animal model are also important in humans. Specifically, Calon et al. (2012) found a list of 172 genes that were associated with gene TGFB in mice, and showed that TGFB plays a crucial role in colon cancer progression. Our goal is to identify what genes within a set of $p = 1000$ genes analyzed in Rossell & Telesca (2017) are associated with TGFB in humans, using data from $n = 262$ patients, by regressing TGFB expression on that of the $p = 1000$ genes. To this end, we run Tran-s-ell0 defining two blocks of variables: one with the 172 genes found in the mouse study, and another with the remaining genes. All variables are standardized to zero mean and unit variance. In this dataset, the runtime of Tran-s-ell0 was 49 seconds.

Table 2 reports the genes selected with Tran-s-ell0, $\hat{S}^{A,sd}$, EBIC, LASSO and SCAD. Compared to the uninformed selector $\hat{S}^{A,sd}$, Tran-s-ell0 selects more genes that were in the mouse list, and fewer genes from outside the list. LASSO and SCAD select a much larger number of genes than Tran-s-ell0, although recall that in our simulations in Section 6.1 these methods exhibited high false discovery rates (see also Bogdan et al. (2015)). The extra discoveries of LASSO and SCAD should hence be considered with caution. Finally, the EBIC selects the same genes from the mouse list as Tran-s-ell0 but also three more genes from outside the list. Recall that in our simulations, EBIC had a slightly larger FDR than our informed selector but also a slightly larger power, for small sample sizes.

To assess the relevance of the extra discoveries of EBIC and the quality of the models selected by EBIC and Tran-s-ell0, we report in Table 3 the associated cross-validated mean squared error (CV-MSE) in estimation. As a measure of variable relevance, we also report pseudo-posterior inclusion probabilities $\pi(\beta_i) = \sum_{M \in \mathcal{M}} NC(M)I(i \in M)$. Compared to

Table 2: Selected genes in each block

	In mouse list		Outside of mouse list	
Method	Number of discoveries	Genes	Number of discoveries	Genes
Tran-s-ell0	4	ESM1, GAS1, HIC1, CILP	0	-
$\hat{S}^{A,sd}$	2	IGFBP3, CILP	2	LGALS1, ARHGEF1
EBIC	4	ESM1, GAS1, HIC1, CILP	3	KCNJ5-AS1, ZBTB46, OXER1
LASSO	22	GAS1, HIC1, IGFBP3, CILP, DNAJB5, ...	55	KCNJ5-AS1, ZBTB46, OXER1, LGALS1, ...
SCAD	11	GAS1, HIC1, IGFBP3, CILP, DNAJB5, ...	5	POLQ, CRIP2, ZBTB46, VSIG4, LGALS1

the EBIC, Tran-s-ell0 leads to lower mean squared error and to higher pseudo-posterior inclusion probabilities for the genes in the mouse list. These findings suggest that there is value in structural transfer learning from mice to humans.

7 Discussion

We studied how variable selection in high-dimensional regression can be improved when one has external information that suggests that variables in some blocks are more likely than those in other blocks to be associated to the response. We first characterized the gains

Table 3: Tran-s-ell0 vs EBIC: cross-validated MSE and pseudo-posterior probabilities

		Pseudo-posterior probabilities			
Method	CV-MSE	ESM1	GAS1	HIC1	CILP
EBIC	0.54	0.55	0.88	0.81	0.95
Tran-s-ell0	0.49	0.74	0.90	0.95	0.99

achievable by an oracle choice of block penalties, which serves as a benchmark for what external information can provide. We then proposed a data-based procedure that learns block penalties from the data and retains the main qualitative benefits of the oracle selector. As an important special case, we analyzed a structural transfer learning setting in which sets of variables selected in source studies are used as external information to guide selection in a target study, yielding a method that is robust to negative transfer.

Several directions remain open. First, it would be interesting to understand whether improvements can be obtained for estimation (not only support recovery) when such external information is available; while our simulations suggest potential gains, we did not develop estimation procedures or theory in this direction. Second, our results focus on linear regression. Extending the analysis to other structural learning problems—such as graphical model selection—could clarify when and how external information can relax the conditions required for consistent structure recovery. Finally, it would be interesting to develop theory for cases where the external information does not simply split variables into blocks, but rather provides a richer description such as continuous measures of variable relevance found in several previous studies.

References

Abba, M. A., Williams, J. P. & Reich, B. J. (2024), 'A bayesian shrinkage estimator for transfer learning'.

Bertsimas, D. & Van Parys, B. (2020), 'Sparse high-dimensional regression', *The Annals of Statistics* **48**(1), 300–323.

Bogdan, M., van den Berg, E., Sabatti, C., Su, W. & Candès, E. J. (2015), 'Slope adaptive variable selection via convex optimization', *The Annals of Applied Statistics* **9**(3), 1103–1140.

Boulesteix, A.-L., De Bin, R., Jiang, X. & Fuchs, M. (2017), 'IPF-LASSO: Integrative L1-penalized regression with penalty factors for prediction based on multi-omics data', *Computational and Mathematical Methods in Medicine* **2017**, 1–14.

Busatto, C. & van de Wiel, M. (2023), 'Informative co-data learning for high-dimensional horseshoe regression'.

Butucea, C., Ndaoud, M., Stepanova, N. A. & Tsybakov, A. B. (2018), 'Variable selection with Hamming loss', *The Annals of Statistics* **46**(5), 1837 – 1875.

Calon, A., Espinet, E., Palomo-Ponce, S., Tauriello, D. V., Iglesias, M., Céspedes, M. V., Sevillano, M., Nadal, C., Jung, P., Zhang, X. H.-F., Byrom, D., Riera, A., Rossell, D., Mangues, R., Massague, J., Sancho, E. & Batlle, E. (2012), 'Dependency of colorectal cancer on a TGF-beta-driven programme in stromal cells for metastasis initiation', *Cancer Cell* **22**(5), 571–584.

Cassese, A., Guindani, M. & Vannucci, M. (2014), 'A Bayesian integrative model for genetical genomics with spatially informed variable selection', *Cancer informatics* **13**, S13784.

Chen, J. & Chen, Z. (2008), 'Extended Bayesian information criteria for model selection with large model spaces', *Biometrika* **95**(3), 759–771.

Chen, T.-H., Chatterjee, N., Landi, M. T. & Shi, J. (2021), 'A penalized regression framework for building polygenic risk models based on summary statistics from genome-wide association studies and incorporating external information', *Journal of the American Statistical Association* **116**(533), 133–143.

Fan, J. & Li, R. (2001), 'Variable selection via nonconcave penalized likelihood and its oracle properties', *Journal of the American Statistical Association* **96**(456), 1348–1360.

Gao, M. & Aragam, B. (2025), 'Optimality and computational barriers in variable selection under dependence'.

Giannone, D., Lenza, M. & Primiceri, G. E. (2021), 'Economic predictions with big data: The illusion of sparsity', *Econometrica* **89**(5), 2409–2437.

Jiang, Y., He, Y. & Zhang, H. (2016), 'Variable selection with prior information for generalized linear models via the prior LASSO method', *Journal of the American Statistical Association* **111**(513), 355–376. PMID: 27217599.

Lai, D., Padilla, O. H. M. & Gu, T. (2024), 'Bayesian transfer learning for enhanced estimation and inference'.

Li, S., Cai, T. T. & Li, H. (2021), 'Transfer learning for high-dimensional linear regression: Prediction, estimation and minimax optimality', *Journal of the Royal Statistical Society Series B: Statistical Methodology* **84**(1), 149–173.

Liang, P. P., Zadeh, A. & Morency, L.-P. (2023), 'Foundations and trends in multimodal machine learning: Principles, challenges, and open questions'.

Liu, Y., Zhang, H., Chen, Q. & Fang, C. (2025), ‘Optimal algorithms in linear regression under covariate shift: On the importance of precondition’.

Rossell, D. (2022), ‘Concentration of posterior model probabilities and normalized L_0 criteria’, *Bayesian Analysis* **17**(2), 565 – 591.

Rossell, D. & Telesca, D. (2017), ‘Non-local priors for high-dimensional estimation’, *Journal of the American Statistical Association* **112**, 254–265.

Schwarz, G. (1978), ‘Estimating the dimension of a model’, *The Annals of Statistics* **6**(2), 461 – 464.

Stingo, F. C., Chen, Y. A., Tadesse, M. G. & Vannucci, M. (2011), ‘Incorporating biological information into linear models: A Bayesian approach to the selection of pathways and genes’, *The Annals of Applied Statistics* **5**(3), 1–24.

Tian, Y. & Feng, Y. (2023), ‘Transfer learning under high-dimensional generalized linear models’, *Journal of the American Statistical Association* **118**(544), 2684–2697.

Tibshirani, R. (1996), ‘Regression shrinkage and selection via the LASSO’, *Journal of the Royal Statistical Society. Series B (Methodological)* **58**(1), 267–288.

van de Wiel, M. A., Te Beest, D. E. & Münch, M. M. (2019), ‘Learning from a lot: Empirical Bayes for high-dimensional model-based prediction’, *Scandinavian Journal of Statistics* **46**(1), 2–25.

van de Wiel, M. A. & van Wieringen, W. N. (2024), ‘Guiding adaptive shrinkage by co-data to improve regression-based prediction and feature selection’.

van Nee, M. M., van de Brug, T. & van de Wiel, M. A. (2023), ‘Fast marginal likelihood estimation of penalties for group-adaptive elastic net’, *Journal of Computational and Graphical Statistics* **32**(3), 950–960. PMID: 38013849.

Velten, B. & Huber, W. (2021), ‘Adaptive penalization in high-dimensional regression and classification with external covariates using variational Bayes’, *Biostatistics* **22**(2), 348–364.

Wainwright, M. J. (2010), ‘Information-theoretic limits on sparsity recovery in the high-dimensional and noisy setting’, *Information Theory, IEEE Transactions on* **55**, 5728 – 5741.

Yang, Y., Wainwright, M. J. & Jordan, M. I. (2016), ‘On the computational complexity of high-dimensional Bayesian variable selection’, *The Annals of Statistics* **44**(6), 2497 – 2532.

Zeng, C., Thomas, D. C. & Lewinger, J. P. (2020), ‘Incorporating prior knowledge into regularized regression’, *Bioinformatics* **37**(4), 514–521.

Zhou, Q., Yang, J., Vats, D., Roberts, G. O. & Rosenthal, J. S. (2022), ‘Dimension-free mixing for high-dimensional Bayesian variable selection’, *Journal of the Royal Statistical Society Series B: Statistical Methodology* **84**(5), 1751–1784.